

IMO(5th) – 1963

First Day

1. Determine all real solution of the equation $\sqrt{x^2 - p} + 2\sqrt{x^2 - 1} = x$, where p is a real number.
2. Find the locus of points in space that are vertices of right angles of which one ray passes through a given point and the other intersects a given segment.
3. Prove that if all the angles of a convex n -gon are equal and the lengths of consecutive edges $a_1, a_2, \dots, a_n$ satisfy $a_1 \geq a_2 \geq \dots \geq a_n$, then $a_1 = a_2 = \dots = a_n$.

Second Day

4. Find all solutions of $x_1, x_2, \dots, x_5$ to the system of equations:

$$\begin{cases} x_5 + x_2 = yx_1 \\ x_1 + x_3 = yx_2 \\ x_2 + x_4 = yx_3 \\ x_3 + x_5 = yx_4 \\ x_4 + x_1 = yx_5 \end{cases}$$

Where y is a real parameter.

5. Prove that $\cos \frac{\pi}{7} - \cos \frac{2\pi}{7} + \cos \frac{3\pi}{7} = \frac{1}{2}$.
6. Five students A, B, C, D, E have taken part in a certain competition. Before the competition, two persons X and Y tried to guess the rankings. X thought that the ranking would be A, B, C, D, E; and Y thought that the ranking would be D, A, E, C, B. At the end it was revealed that X didn't guess correctly any ranking of the participants, and moreover, didn't guess any of the ordering of pairs of consecutive participants. On the other hand, Y guessed the correct rankings of two participants and the correct ordering of two pairs of consecutive participants. Determine the ranking of the participants.

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